Semiparametric Modelling and Forecasting Economic Time Series

The aim of this project is to apply nonparametric regression, ARMA models as well as other relevant statistical tools for analysing some real economic time series. Also short or middle term forecasting will be considered. The considered data examples are e.g. those of the national GDP, unemployment rates, some most important macroeconomic indices, in Germany or in your country, if you are an international student. Considered are mainly monthly time series. You can also choose a few quarterly time series as examples.

The basic model for this project is the traditional additive model for economic time series with an trend component for the business cycle, a seasonal component for yearly periodic change and a stationary error process. It is assumed that the first two components are deterministic and the last one follows an ARMA model. Nonparametric techniques, in particular using locally weighted regression, are required to model the deterministic components. The residuals will then be modelled by an ARMA model. So the whole model is called a semiparametric time series model. This approach is a recent extension of the forth version of the Berlin-Method (BV4, Berliner Verfahren 4), the computer package for decomposing seasonal time series currently used by the Statistisches Bundesamt Deutschland (DeStatis) and the Deutsches Institut für Wirtschaftsforschung (DIW Berlin). The calculation will be mainly done in R. The main packages you need will be provided.

The selection of different ARMA models using the BIC (Bayesian Information Criterion) will also be discussed. Another focus is to apply the fitted model for forecasting. The question how possible outliers can be dealt with should also be discussed. This idea will then be used to analyse the possible effect the financial crisis on the business cycle, seasonality and on the error process as well.

Another point is that through this MA project you can know well what empirical economic research is and who are working on empirical economic research in Germany.

Selected references:

1. Cleveland, R.B., Cleveland, W.S., McRae, I.E., Terpenning, I. (1990). STL: A seasonal-trend decomposition procedure based on LOWESS (with discussion). J. Official Statistics, 6, No. 1, 3-73.
2. Fan, J. and Gijbels, I. (1996). Local Polynomial Modeling and its Applications. Chapman & Hall, London.
3. Feng (1999). *Kernel and Locally Weighted Regression*. Verlag f¨ur Wissenschaft und Forschung, Berlin.
4. Feng, Y. and Heiler, S. (2000). Eine robuste datengesteuerte Version des Berliner-Verfahrens (in German). *Wirtschaft und Statistik*, 10/2000, 786-795.
5. Feng, Y. (2003). Semiparametric modelling of seasonal time series. *Bulletin of the International Statistical Institute*, Volume LX, Book 1, pp. 334-335, Berlin.
6. Grosser, H. (2007). Trends und Kennzahlen. *Münchner Statistik*, 21-30.
7. Härdle (1990), *Applied nonparametric Regression*. Cambridge University Press, New York.
8. Heiler, S. and Feng, Y. (2000). Data-driven decomposition of seasonal time series. *Journal of Statistical Planning and Inference*. 91, 351-363.
9. Speth, H.-T. (2004). Komponentenzerlegung und Saisonbereinigung ökonomischer Zeitreihen mit dem Verfahren BV4.1. Statistisches Bundesamt.